## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 1 of 5 · Question 3 part b · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 3 | (a) (i) | • time in minutes;<br>• 0-10, 11-20, 21-30, 31-40,41-50,51-60; | | 2 |
| | (ii) | • 75 + 72 + 68; plus/minus 2 for each one<br>• 215 $cm^3$; | ECF | 2 |
| | (iii) | • rapid reaction/a lot/large amount of gas produced, at start;<br>• rate slows/less gas (for next 10 minutes);<br>• drops a lot/little gas produced (after 40 minutes); | | 3 |
| | (iv) | any three from<br>• at start plenty of substrate/many E-S complexes formed;<br>• substrate used up/less substrate available;<br>• reaction slows/less gas/product, produced/less enzyme action;<br>• most substrate used at end so little gas produced; | | 3 |
| | (b) | • less gas produced/reaction lower/less;<br>• because active sites blocked; | | 2 |

Total 12

## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 1 of 5 · Question 3 part b · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 2 of 5 · Question 5 part b.iii · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 5 | (a) | $4^3$;<br>= 64; | Full marks for correct answer | 2 |
| | (b) (i) | • change in order/sequence;<br>• of bases;<br>• different protein may be formed; | | 3 |
| | (ii) | uv light/λ rays/ionising radiation/X-rays; | accept any suitable example | 1 |
| | (iii) | • can't catalyse reaction/slows reaction/activity;<br>• because can't form E-S complexes/substrate can't bind with active site/active site not complementary; | | 2 |
| | (iv) | doesn't affect working of enzyme/mutation does not change amino acid/doesn't affect structure/shape of enzyme/different coding for same amino acid; | | 1 |

Total 9 marks

# Type MS · Explain how inhibitors or a mutation change enzyme activity

## Answer 2 of 5 · Question 5 part b.iii · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 3 of 5 · Question 9 part c · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 9 | (a) (i) | • reference to platelets;<br>• fibrinogen converted to fibrin;<br>• (fibrin) mesh/crosslinks/fibres formed; | | 3 |
| | (ii) | • reduced blood flow to heart/cardiac muscle / tissue/cells;<br>• less oxygen/glucose delivered;<br>• less (aerobic) respiration/energy released (by cardiac tissue); | | 3 |
| | (b)(i) | size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw; | Allow valid alternatives | 1 |
| | (ii) | use water without aspirin dissolved; | | 1 |
| | (iii) | • the more tablets dissolved the more solution collected;<br>• the more tablets dissolved the quicker the solution is collected;<br>• aspirin dissolves/acts on blood clot;<br>• size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets; | | Max 3 |
| | (c) | Any four from:<br>• aspirin binds to enzyme/aspirin competes (with substrate) for active site;<br>• active site of enzyme changed/blocked;<br>• less/no substrate binds (to enzyme);<br>• reaction involving release of chemicals reduced/stopped;<br>• platelets do not stick together; | | Max 4 |

**Total question 9 = 15 marks**

## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 3 of 5 · Question 9 part c · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 4 of 5 · Question 2 part b.iii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) (i) | | • a tube containing inhibitor and substrate;<br>• but no enzyme; | | 2 |
| | (ii) | any three from<br>• concentration of enzyme;<br>• concentration of inhibitor;<br>• temperature;<br>• pH;<br>• same filter in colorimeter; | | 3 |
| (b) (i) | | axes labelled with units;<br>axes correct way round;<br>plots; | | 3 |
| | (ii) | suitable curve; | | 1 |
| | (iii) | similar shaped curve;<br>reaching maximum sooner; | | 2 |
| | (c) | same maximum never reached;<br>despite increase in substrate concentration; | | 2 |

Total 13 marks

## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 4 of 5 · Question 2 part b.iii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 5 of 5 · Question 2 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | • a tube containing inhibitor and substrate;<br>• but no enzyme; | | 2 |
| | (ii) | any three from<br>• concentration of enzyme;<br>• concentration of inhibitor;<br>• temperature;<br>• pH;<br>• same filter in colorimeter; | | 3 |
| (b) | (i) | axes labelled with units;<br>axes correct way round;<br>plots; | | 3 |
| | (ii) | suitable curve; | | 1 |
| | (iii) | similar shaped curve;<br>reaching maximum sooner; | | 2 |
| | (c) | same maximum never reached;<br>despite increase in substrate concentration; | | 2 |

Total 13 marks

## Type MS · Explain how inhibitors or a mutation change enzyme activity

### Answer 5 of 5 · Question 2 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|